MAITELE ORWELL MAKUNGO

+27-61 495 0245

maitmak8@gmail.com

SUMMARY

Final-year Mechatronics Engineering student with strong analytical and problem-solving skills and practical experience applying Python, machine learning and data analysis to solve real-world problems. Passionate about continuous improvement. Experience delivering technical reports. Seeking to apply engineering thinking to improve operational efficiency within financial services.

EDUCATION

Noordwyk Secondary School

NSC
2016 – 2020
DUX Scholar award winner in Matric

University of Cape Town

Mechatronics Engineering
2021 – present

SKILLS

- Python
- MATLAB
- C++
- Java
- Machine Learning

CERTIFICATIONS

- Multiple Matlab certifications that can be found on LinkedIn. Examples include Machine learning , Deep learning and Reinforcement Learning.

PROFESSIONAL EXPERIENCE

- As a part of the Trading club and the Innovation analyst team, we did a portfolio report on a Spanish renewable energy company named Iberdrola (IBE, MC).
- The report contained the company description, market and industry overview, competition analysis, financial and risk analysis, discounted cash flow evaluation preliminaries, discounted cash flow evaluations, security market line and ESG analysis.
- Was part of Trading club's European equities team. Wrote a report for Deutsche bank in 2025 recommending a buy and it increased by 33% that year.

Personal Projects

- Designed automated decision-making systems using MQL5
- Applied signal processing to analyse financial data.
- Automated trading workflows and evaluated performance using quantitative metrics.
- Automation & Embedded Systems
- Developed embedded sensor systems using Arduino.

Extra-curricular activities

- Part of UCT's Investment Society (largest in Africa).
- Part of the Trading club which is a subset of the Investment society.
- Teach high school learners Mathematics through Teachout, a club at UCT.

Engineering Intern (non-finance related)

Polyoak Packaging

- Implemented a pneumatic cylinder circuit that oscillates.
- Aided the Polyoak team with maintenance on their industrial machines and got to learn about their designs and internal processes.
- Provided a report on how they could potentially use machine learning in their maintenance processes

Relevant University Coursework (all the course codes can be seen in the attached transcript)

- Digital Signal Processing (EEE4114F): Designed and evaluated a predictive machine learning solution in Python for room occupancy estimation, applying data preprocessing, feature analysis, model evaluation, and performance optimisation to support data-driven decision making, achieving 99.84% validation accuracy.
- CSC1015F: Coded 10 python assignments that introduced python concepts such as files, for-loops and the like.
- CSC1016S: Coded 10 Java assignments that covered basic Java concepts such as classes and GUI.
- Embedded systems I and II(EEE2046S, EEE3096S): Used C++ to write various programs to a microcontroller, such as testing the onboard pulse-width modulator by attaching the microcontroller to an amplifier circuit and then to a speaker. Then I wrote sounds of different wave-shapes (Sinusoids, Triangular and Square) to the micro-controller to test its quality. Also conducted throughput tests and other benchmarking tests.